

Dimensionality Reduction and Feature Selection for Network Intrusion Detection on Apache Spark ML

...with a Leakage-Aware Evaluation Pipeline

Thai Nguyen Vu¹ Dung Van Bui² Nhut Tri Do² Van Du Nguyen³

FAIR'2026

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Seven minutes on a single claim: on CICIDS2017, a near-perfect F1 is usually a property of the evaluation pipeline, not of the model.

I. PROBLEM AND CONTRIBUTIONS

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└ I. PROBLEM AND CONTRIBUTIONS

I. PROBLEM AND
CONTRIBUTIONS

A near-perfect F1 on CICIDS2017 is a warning sign

Two problems, usually treated separately

- **Cost.** ≈ 80 features per flow.
- **Trust.** Near-perfect binary F1 usually signals **data leakage**.

Two concrete leakage sources

1. `destination_port` encodes the *attacked service*.
2. **Feature-level duplicates** survive full-row dedup — the same sample lands in train *and* test.

This paper

Compare four reduction strategies over the same eight Spark ML classifiers — but first make the *evaluation pipeline* trustworthy.

The contribution is the protocol

Not “which method wins” but **how to measure so the answer means something**.

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I. PROBLEM AND CONTRIBUTIONS

A near-perfect F1 on CICIDS2017 is a warning sign

Frame the talk as a measurement paper. The port gives the model a port-to-label shortcut; 80 features are expensive to train, to score and to fit on an edge device. Both leakage sources are removed before any split.

A near-perfect F1 on CICIDS2017 is a warning sign

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- **Trust.** Near-perfect binary F1 usually signals **data leakage**.

Two concrete leakage sources

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This paper

Compare four reduction strategies over the same eight Spark ML classifiers — but first make the evaluation pipeline trustworthy.

The contribution is the protocol

Not “which method wins” but **how to measure so the answer means something**.

1. A **uniform comparison** of RF Importance, SHAP and PCA across the same 8 Spark ML algorithms + 2 ensembles on CICIDS2017.
2. SHAP contrasted **directly** against RF Importance on the same training set.
3. A **leakage-aware, statistically tested** pipeline: leak removal, deterministic feature-level dedup, validation-based selection, exact paired permutation test with Nadeau–Bengio correction.
4. A **per-attack-type multiclass** evaluation — where the methods genuinely differ.
5. An **edge-aware** recommendation: F1 *and* feature count *and* model size, for a Jetson Orin Nano Super (deployment reported in a companion systems paper).

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I. PROBLEM AND CONTRIBUTIONS

Contributions

Five contributions; three, four and five are the ones rarely done together.

Contributions
1. A uniform comparison of RF Importance, SHAP and PCA across the same 8 Spark ML algorithms + 2 ensembles on CICIDS2017.
2. SHAP contrasted directly against RF Importance on the same training set.
3. A leakage-aware, statistically tested pipeline: leak removal, deterministic feature-level dedup, validation-based selection, exact paired permutation test with Nadeau–Bengio correction.
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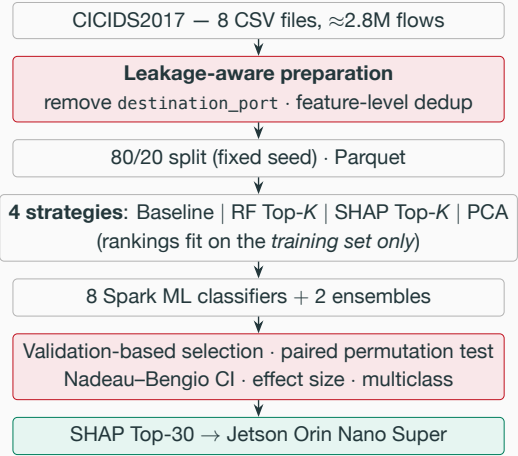
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└ II. LEAKAGE-AWARE PIPELINE

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The leakage-aware pipeline



Removals happen before the split

- Feature-identical groups with **conflicting labels** are dropped *entirely*: **270 groups, 44,260 rows**.
- Result: 77 features, 1.79M / 0.45M train/test flows.

Design fixed a priori

- Exploration (K sweep) separated from confirmation.
- Hyper-parameters shared across all methods.
- The test set enters **no** selection step.

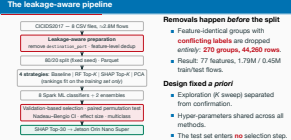
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II. LEAKAGE-AWARE PIPELINE

└ The leakage-aware pipeline

Deterministic deduplication is the part people get wrong. Conflicting groups are removed wholesale, so the result does not depend on how Spark partitions the data.



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└ III. RESULTS AND CONCLUSION

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Result 1: 60% of the features are removable

Method	Model	F1	Train (s)	Feat.
Baseline	XGBoost	0.9971	4615.5	77
SHAP Top-30	XGBoost	0.9970	1226.1	30
PCA k=40	XGBoost	0.9939	1444.8	40
RF Top-30	XGBoost	0.9922	1295.1	30

Algorithm	RF Top-30	SHAP Top-30	Δ
Random Forest	0.9879	0.9955	+0.77%
XGBoost	0.9922	0.9970	+0.48%
GBT	0.9844	0.9918	+0.75%
Decision Tree	0.9835	0.9948	+1.15%

- SHAP Top-30 keeps baseline F1 while cutting **≈60% of features** and **≈73% of training time**.
- At *equal* feature count, SHAP beats RF Importance on every tree-based algorithm.
- PCA uses *k*=40 because it *extracts* rather than *selects* — each component carries only part of the variance. Cheaper than the baseline, but not explainable.

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III. RESULTS AND CONCLUSION

Result 1: 60% of the features are removable

Two tables, one message: SHAP Top-30 is the operating point. Same accuracy, 60 percent fewer features, and it is explainable, which matters for the SOC analyst and for the edge board downstream.

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Result 2: testing the difference instead of asserting it

Method	Model	F1 (mean ± CI95)
Baseline	XGBoost	0.99734 ± 0.00007
SHAP Top-30	XGBoost	0.99720 ± 0.00009
paired permutation $p = 0.001$, $d = 1.94$		

Port ablation (RF)	F1	Recall (Attack)
Keep port	0.9965	0.9983
Remove port	0.9955	0.9961

Significant *only at the resolution floor*, gap ≈ 0.0001 : **practically equivalent**, so explainability and cost decide. After dedup the port’s isolated effect is small — duplicates were the dominant leak.

- $N=6$ paired *resamplings*: variance reflects data sampling, not just seeds.
- Two-sided permutation test, **exact enumeration** of all 2^6 sign flips: smallest reachable p is $2/2^6$, so $N \geq 6$ is *required*.
- Overlapping resamplings break t -independence → we use the **Nadeau–Bengio corrected CI**.

Nadeau & Bengio, *Inference for the Generalization Error*, Machine Learning 52(3), 2003.

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III. RESULTS AND CONCLUSION

Result 2: testing the difference instead of asserting it

Variance reflects data sampling, not just model initialization. The test is two-sided, and overlapping resamplings break t -independence. Small p , tiny effect in absolute terms: say plainly that we treat them as equivalent rather than claiming a winner.

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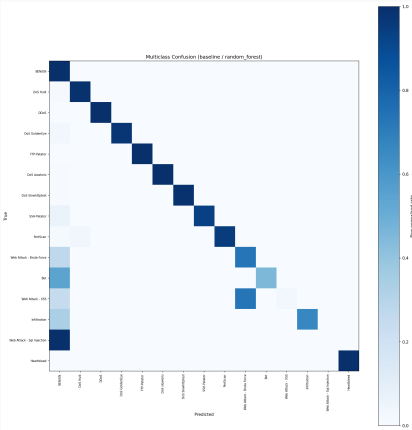
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Result 3: binary F1 hides the rare classes



Class	Support	Recall	F1
Benign	380,119	0.9997	0.9989
DoS Hulk	34,446	0.9904	0.9947
Web – Brute Force	311	0.7299	0.7323
Bot	290	0.4552	0.6256
Web – XSS	111	0.0270	0.0526
Web – SQL Injection	6	0.0000	0.0000
weighted-F1 = 0.998		macro-F1 = 0.806	

- Two failure modes, read by destination:
- **Mislabeled but caught:** 81/111 XSS land on Web Brute Force — **76% still flagged** at the binary level.
 - **Leaking into Benign** (dangerous): all 6 SQL Injection, **54% of Bot**, 27% of Brute Force.

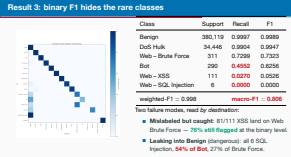
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III. RESULTS AND CONCLUSION

Result 3: binary F1 hides the rare classes

Weighted F1 0.998 versus macro F1 0.806. Three causes: extreme imbalance; removing the port hits web attacks hardest, because without the port-80 shortcut short HTTP flows look like browsing, so the 0.027 recall reported here replaces a leakage-inflated one; and Bot C&C beaconing resembles background traffic.



Result 4: the scores do not survive a different testbed

Train → Test	F1	AUC-PR
2017 → 2017	0.9953	0.9997
2017 → 2018	0.029–0.085 0.185–0.301	0.4849
2018 → 2018	0.9246	0.9679
2018 → 2017	0.0544	0.1926

Mean of 18 runs, shared leakage-free features, ~2.39M flows.

- In-domain stable *both* ways — cross-dataset AUC-PR **1.00→0.48** and **0.97→0.19**.
- Recall 0.073 / 0.030 at precision 0.772 / 0.272: the model **fails to recognize** the other testbed’s attacks.
- 18 runs, **two disjoint regimes**: 11 at 0.029–0.085, 7 at 0.185–0.301, nothing between. AUC-PR identical in both.
- Causes: CICFlowMeter version, network configuration, attack tooling.

Near-perfect in-domain scores — *even leakage-free ones* — do not certify deployability under distribution shift.

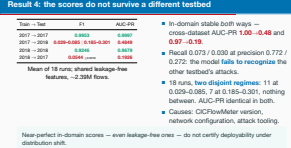
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III. RESULTS AND CONCLUSION

Result 4: the scores do not survive a different testbed

Do not soften this. In-domain 0.99, cross-dataset AUC-PR 0.50 / 0.19. If asked about the wide F1 interval: at this recall the metric is fragile to where the threshold falls, so we average F1 over runs and judge the gap on AUC-PR. Consistent with prior cross-dataset IDS studies, and the reason domain adaptation is a prerequisite rather than a nice-to-have.



What we showed

- SHAP Top-30 retains baseline F1 with **≈60% fewer features**.
- Baseline vs. SHAP Top-30 are **practically equivalent**.
- **macro-F1 0.806** against weighted-F1 0.998.
- Cross-dataset AUC-PR **collapses to 0.50 / 0.19**.

The contribution

- **How to evaluate reliably**, as much as which method wins.

Next

- Full CSE-CIC-IDS2018; newer sets (RoEduNet, CIC-IoT).
- Does the SHAP Top-30 *set itself* transfer?
- Targeted imbalance handling; temporal features for beaconing.

Thank you — questions?

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III. RESULTS AND CONCLUSION

Conclusion

Close on the protocol, not the leaderboard.

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Key references

[1] Sharafaldin, I., Lashkari, A. H. & Ghorbani, A. A. *Toward generating a new intrusion detection dataset and intrusion traffic characterization*. ICISSP 2018. **CICIDS2017**

[2] Lanvin, M. et al. *Errors in the CICIDS2017 Dataset and the Significant Differences in Detection Performances It Makes*. CRiSIS 2022. **label conflicts**

[3] Arp, D. et al. *Dos and Don'ts of Machine Learning in Computer Security*. USENIX Security 2022. **leakage pitfalls**

[4] Lundberg, S. M. & Lee, S.-I. *A unified approach to interpreting model predictions*. NeurIPS 2017. **SHAP**

[5] Nadeau, C. & Bengio, Y. *Inference for the Generalization Error*. Machine Learning 52(3), 2003. **corrected CI**

[6] D'hooge, L. et al. *Inter-dataset generalization strength of supervised machine learning methods for intrusion detection*. J. Information Security and Applications 54, 2020. **cross-dataset**

[7] Zaharia, M. et al. *Spark: Cluster computing with working sets*. HotCloud 2010. **Spark ML**

[8] CIC & CSE. *CSE-CIC-IDS2018 dataset*. 2018. **cross-dataset target**

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III. RESULTS AND CONCLUSION

Key references

Not presented — shown while taking questions, so sources are on screen if the audience asks. Full list is in the paper.

Key references	
[1]	Sharafaldin, I., Lashkari, A. H. & Ghorbani, A. A. <i>Toward generating a new intrusion detection dataset and intrusion traffic characterization</i> . ICISSP 2018. CICIDS2017
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[6]	D'hooge, L. et al. <i>Inter-dataset generalization strength of supervised machine learning methods for intrusion detection</i> . J. Information Security and Applications 54, 2020. cross-dataset
[7]	Zaharia, M. et al. <i>Spark: Cluster computing with working sets</i> . HotCloud 2010. Spark ML
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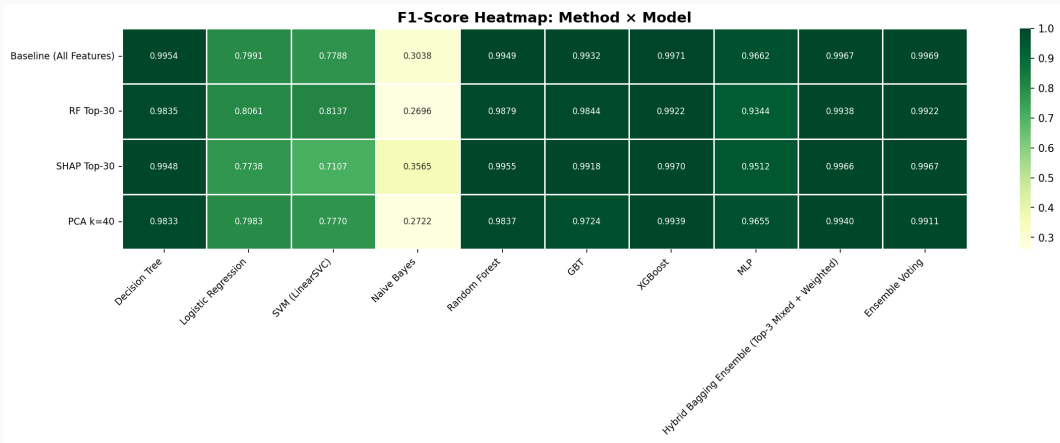
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Backup slides

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Backup: F1 heatmap, method × algorithm

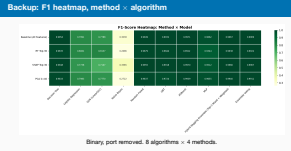


Binary, port removed, 3 algorithms × 4 methods.

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Backup: F1 heatmap, method × algorithm



Backup: hyper-parameters, fixed a priori

Model	Main hyper-parameters
Decision Tree	maxDepth=15, minInstancesPerNode=5, impurity=entropy
Logistic Regression	maxIter=200, regParam=0.001, L2, threshold=0.5
SVM (LinearSVC)	maxIter=200, regParam=0.001
Naive Bayes	gaussian, smoothing=1.0
Random Forest	numTrees=200, maxDepth=15, featureSubset=sqrt
GBT	maxIter=150, maxDepth=6, stepSize=0.05, subsample=0.8
XGBoost	n_estimators=300, max_depth=8, lr=0.05, subsample=0.8, colsample=0.8
MLP	layers=[d,64,32,2], maxIter=150, stepSize=0.01

Shared across all reduction methods, so differences reflect the *method*, not per-model tuning.
Class weights for 6 models, scale_pos_weight for XGBoost; MLP unweighted (Spark ML has no sample weights for it) — stated in the limitations. LightGBM excluded: its SynapseML native library is x86_64-only, and the deployment target is ARM64.

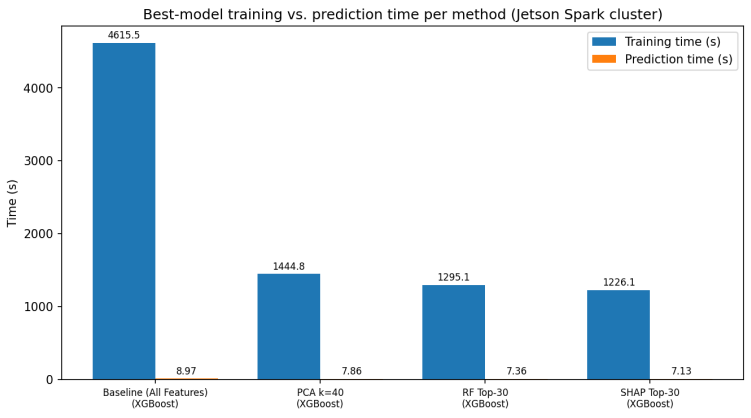
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Backup: training and prediction cost



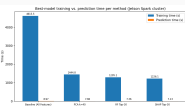
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Backup: threats to validity

- **Internal.** RF/SHAP rankings come from the original training set and stay fixed across resamplings — consistent with the a-priori design, but a possible leak in the ranking step. Re-ranking inside two of the six resamplings moved F1 by $\leq 7.4 \times 10^{-5}$, and 25–26 of 30 features were reselected. MLP carries the unweighted-training caveat.
- **External.** CICIDS2017 may not reflect current traffic, and benchmark NIDS datasets have documented quality limitations. The cross-dataset conclusion rests on a *stratified sample* of CSE-CIC-IDS2018. Evasive traffic untested. Feature-level dedup itself shifts the benign/attack distribution.
- **Statistical.** $N=6$ gives low power: two-sided exact enumeration floors p at $2/2^N$, so $N=6$ just reaches 0.05. Resamplings overlap, violating t -independence — hence Nadeau–Bengio as the primary basis. Nested CV with larger N is the next reinforcement.

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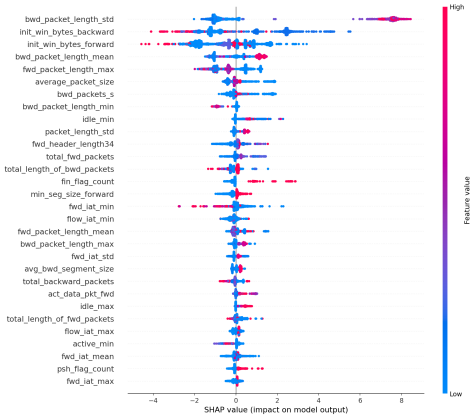
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Backup: SHAP feature attribution



SHAP is computed on a *class-stratified* sample so both benign and attack traffic are represented in the ranking.

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Backup: SHAP feature attribution

